Lecture 10: Human Reproduction

- Human reproduction & puberty
- Female reproductive anatomy
- Female hormone cycles
- Male reproductive anatomy
- Preventing pregnancy
- Infections transmitted through sexual interaction

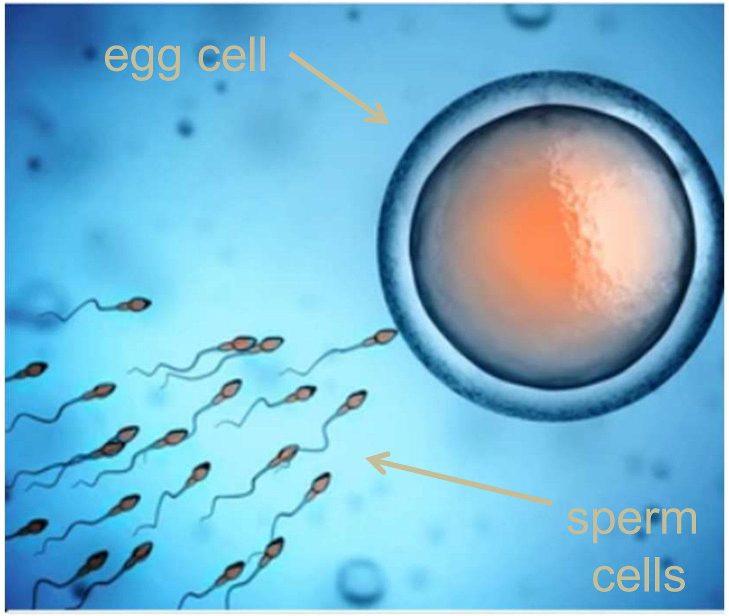
Corresponds with OpenStax Biology 2e Chapter 43

Human reproduction & puberty

- *Remember*: organs called **gonads** produce haploid sperm or eggs through meiosis
  - Gonads = **testes** or **ovaries**

- *Remember*: we usually describe two sexes
  - Females: two X chromosomes, development of ovaries which produce eggs
  - Males: an X & Y chromosome, development of testes which produce sperm
    - *Not everyone fits into these two categories*
    - *Sex is different than gender*

- Ovaries produce **eggs**: large, nonmotile haploid cells containing food reserves that can nourish an embryo



- Testes produce **sperm**: small, motile haploid cells with no food reserves

- **Fertilization** = the union of haploid sperm & egg, which forms a diploid embryo
  - Fertilized egg undergoes mitosis to produce offspring

    → *reproduction*

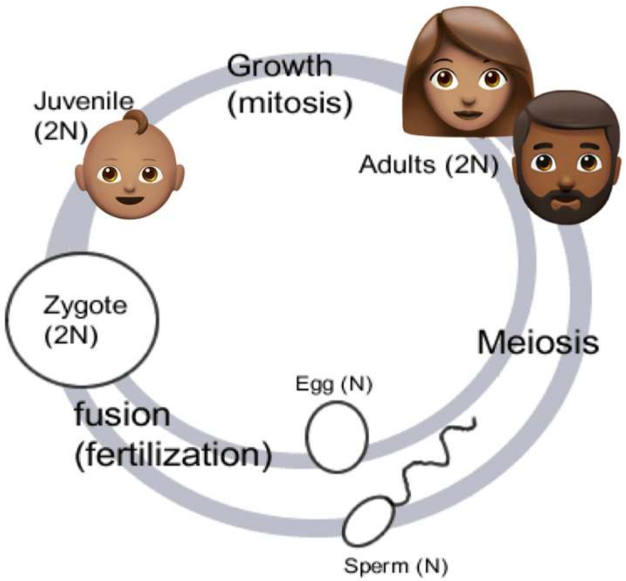
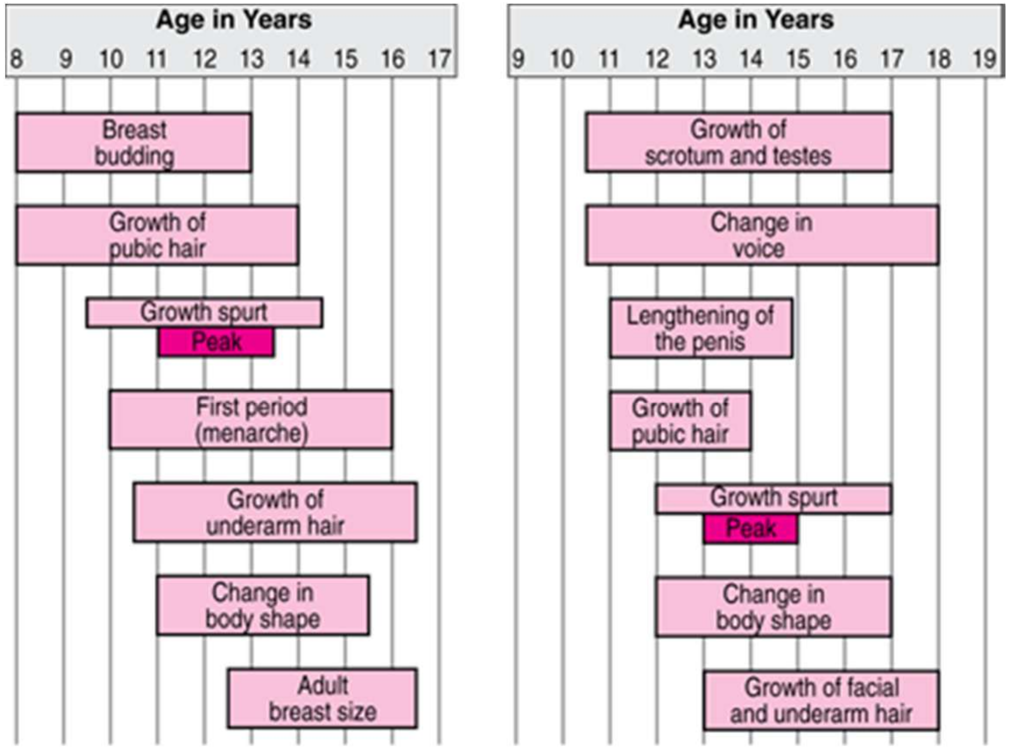


Figure: https://www.biologycorner.com/APbiology/inheritance/10-1_meiosis.html

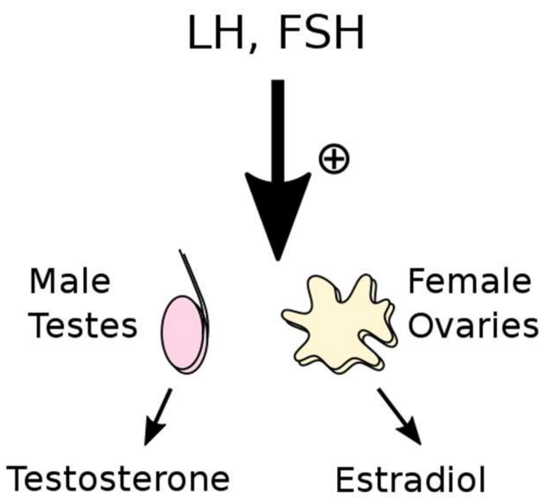
Figure: https://www.shutterstock.com/image-illustration/sperm-egg-cell-128623430

- The ability to reproduce begins at puberty
  - **Puberty** = a stage of development with rapid growth & the appearance of *secondary sexual characteristics*
  - Puberty generally begins in the early teens, but can start as early as age 8 or as late as age 15

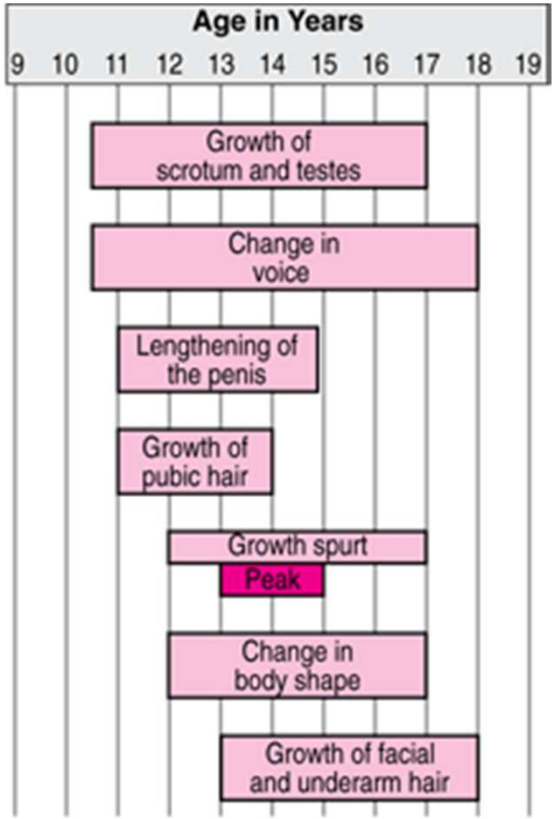


Average changes for females
Average changes for males

- Brain maturation at a certain age of development causes the release of **luteinizing hormone (LH)** & **follicle-stimulating hormone (FSH)**
  - LH & FSH stimulate:
    - testes to produce the sex hormone **testosterone**
    - ovaries to produce the sex hormone **estrogen** (the main form is estradiol)
      - Remember: testosterone & estrogen = sterols (lipids)

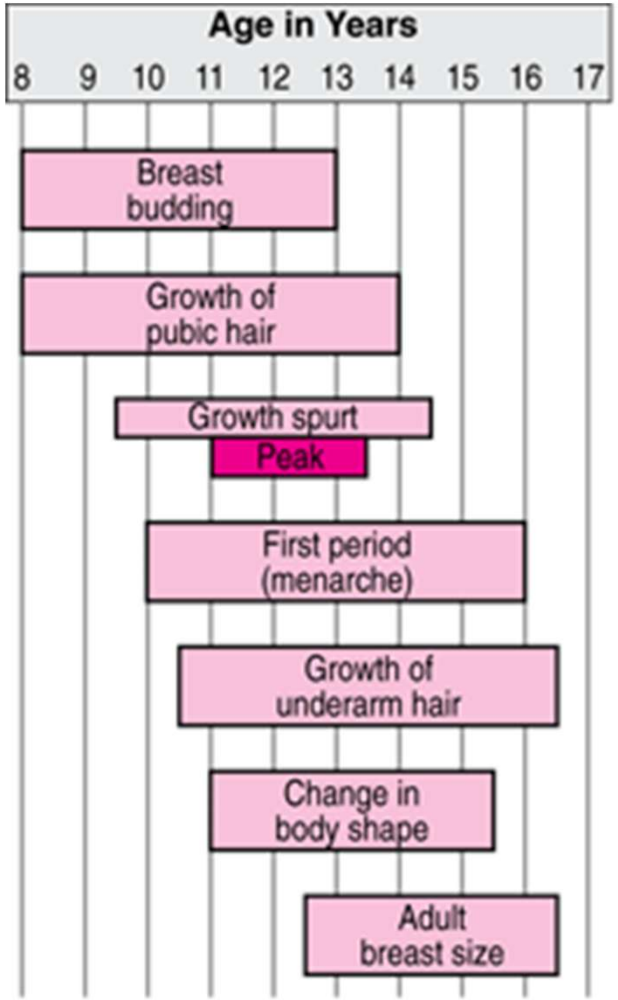


- <u>Secondary sexual characteristics resulting from an increase in **testosterone**:</u>
  - Pubic, underarm, & facial hair appears
  - Penis & testes enlarge
  - Larynx enlarges, deepening the voice
  - Muscular development increases
  - Courtship behaviors can appear



Average changes for males

- <u>Secondary sexual characteristics resulting from an increase in **estrogen**:</u>
  - – Breasts enlarge
  - – Hips become wider
  - – Pubic & underarm hair appears
  - – Menstruation begins
  - – Courtship behaviors can appear



Average changes for females

Female reproductive anatomy

- The female reproductive system
  - Almost entirely contained within the abdominal cavity
  - Consists of structures that produce eggs & hormones, accept sperm, conduct sperm to the egg, & nourish a developing embryo

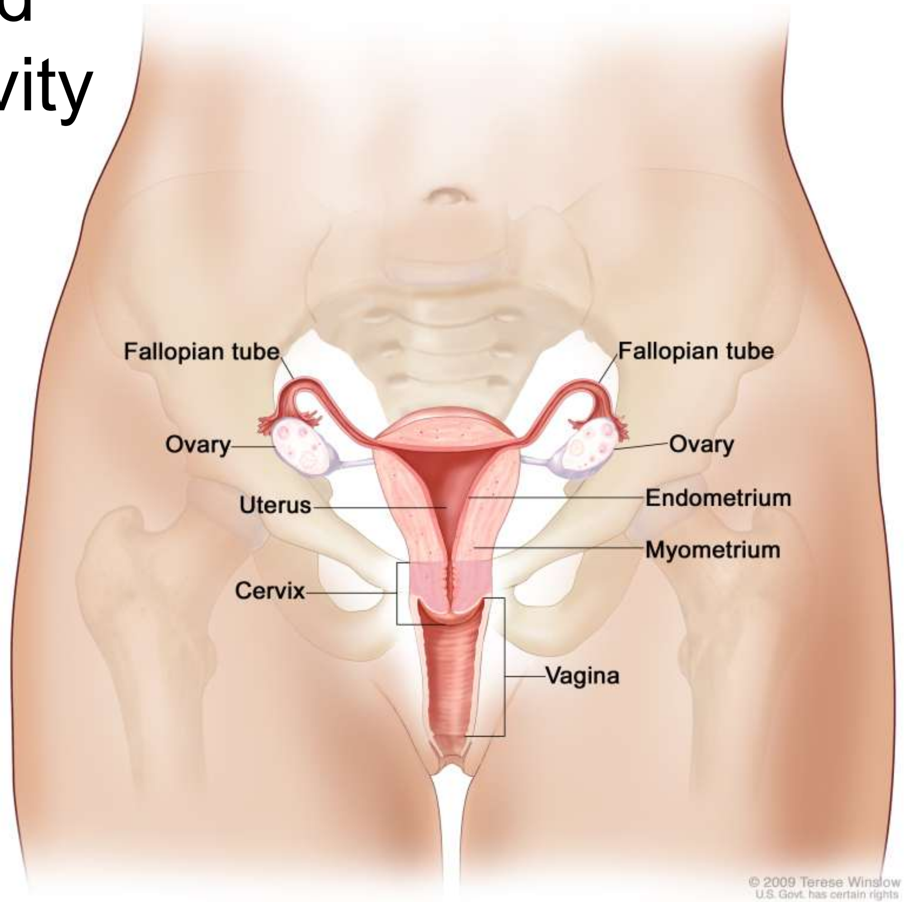
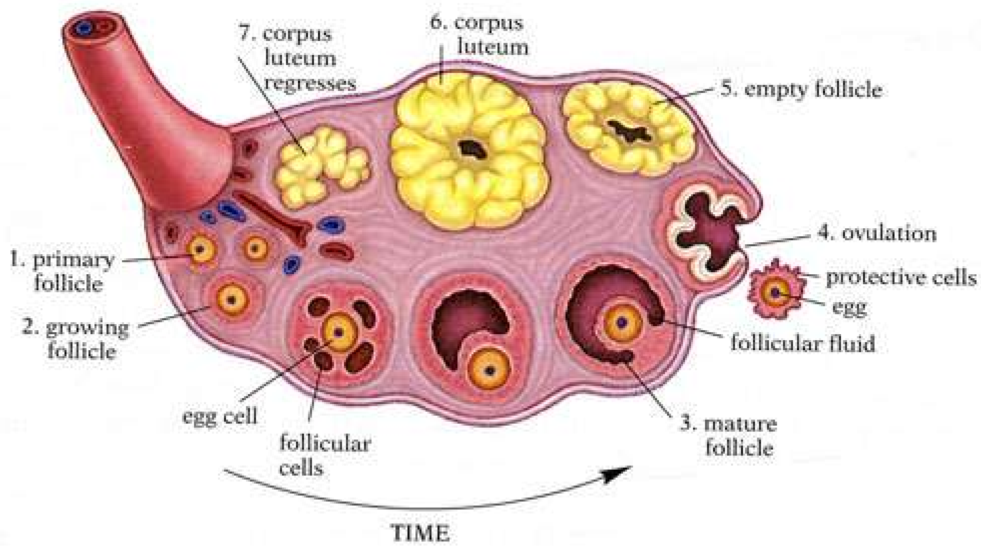


Figure: https://www.ncbi.nlm.nih.gov/books/NBK65829/

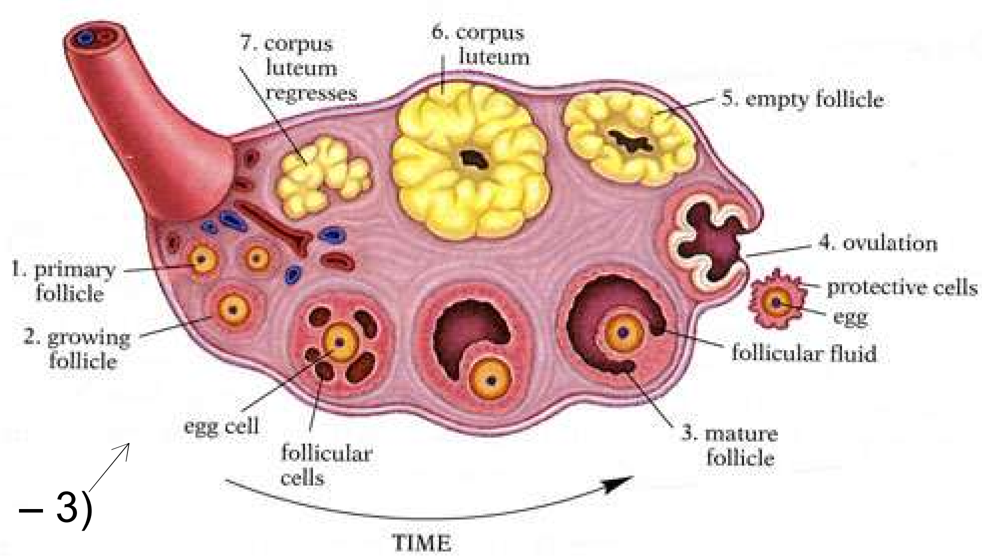
Human female reproductive anatomy

Structure	Function in Reproduction
Ovaries	produce eggs, estrogen, & progesterone
Fallopian tube	site of fertilization, conducts egg to uterus
Fimbriae	cilia that sweep egg into & along uterine tube
Uterus: Endometrium	lining of uterus, site of implantation, nourishes embryo, forms placenta
Uterus: Myometrium	muscular wall, large contractions during labor, small contractions during menstruation
Cervix	nearly closes off uterus holding fetus in, expands during contractions
Vagina	accepts semen, birth canal

- <u>Egg production in the ovaries:</u>
  - Each egg starts meiosis <u>before birth</u>, but stops before completion
    - Females are born with their lifetime supply of eggs (~1 to 2 million) – no new ones are generated later in life
    - Eggs complete meiosis starting at puberty, but only one egg at a time



- <u>Egg production in the ovaries:</u>
  - Surrounding each egg is a layer of smaller cells
    - Together, the egg & these cells make up a **follicle**
  - Roughly every month after puberty, the hormonal changes of the menstrual cycle stimulate the development of a new follicle
    - As the follicle matures, it becomes larger & fills with fluid (steps 1 – 3)



- Egg production in the ovaries:
  - **Ovulation** occurs when the egg & some accompanying follicle cells erupt through the surface of the ovary (step 4 in figure)
  - The rest of the follicle cells remain in the ovary, enlarging & forming a temporary gland called the **corpus luteum** (steps 5 – 7)

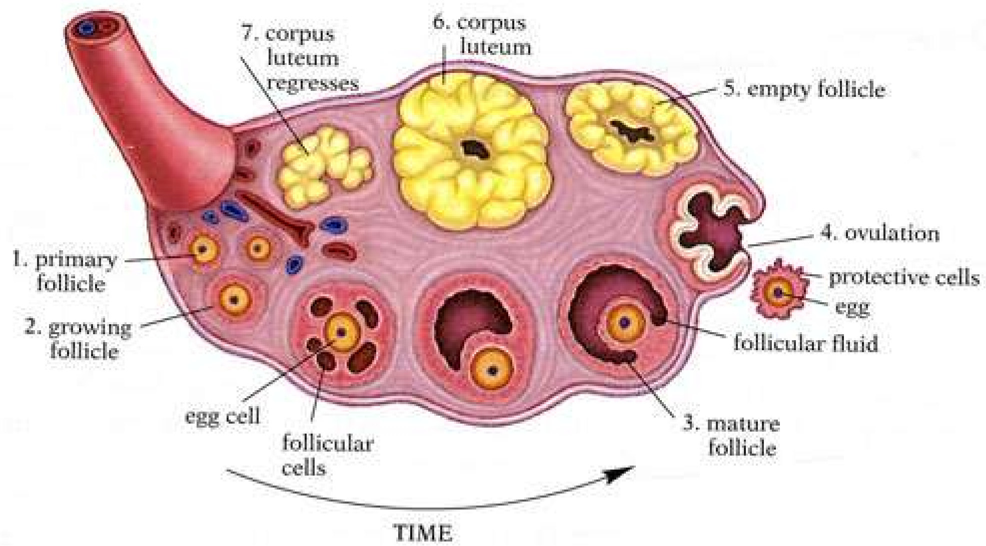


Figure: https://commons.wikimedia.org/wiki/File:Anatomy_of_the_ovaries.jpg

- <u>Egg production in the ovaries:</u>
  - The corpus luteum secretes both estrogen & a 2nd hormone called **progesterone**
    - Estrogen & progesterone stimulate the development of the uterine lining, called the **endometrium**, & help control the menstrual cycle
  - If pregnancy does not occur, the corpus luteum degenerates a few days later (step 7)

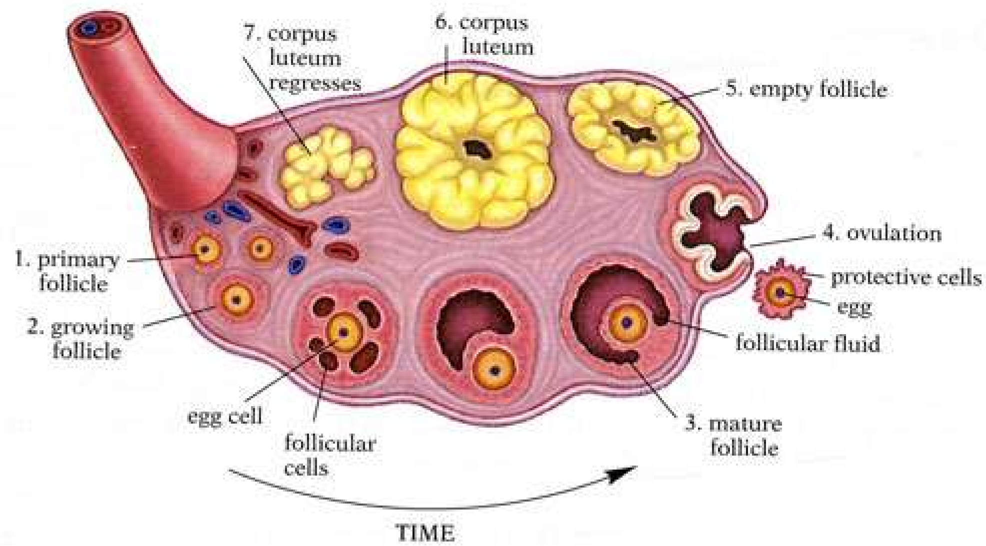


Figure: https://commons.wikimedia.org/wiki/File:Anatomy_of_the_ovaries.jpg

- Each ovary nestles within the open end of a **uterine tube** (also called the oviduct or Fallopian tube)
  - Uterine tube is fringed with cilia called **fimbriae**
    - *Remember*: <u>cilia</u> move things past cells
    - Fimbriae create a current that sweeps the newly ovulated egg into the uterine tube
  - There, the egg may encounter sperm & be fertilized
  - Fimbriae continue to sweep the egg (fertilized or not) down the uterine tube & into the **uterus**

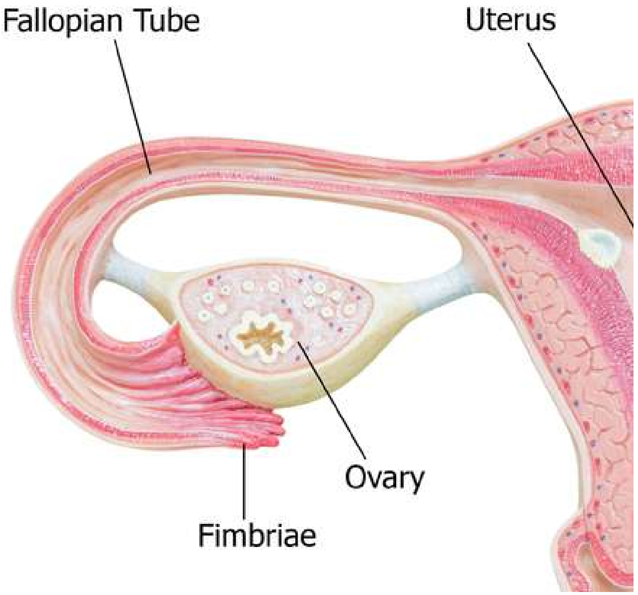


Figure: https://www.shutterstock.com/image-illustration/female-reproductive-system-image-diagram-243154639

- The **endometrium** = the inner lining of the uterus, & is richly supplied with blood vessels & nutrients
  - If the egg is fertilized & successfully implants here, this will form the **placenta**
    - placenta = structure that transfers O^2, nutrients, & wastes between parent & embryo

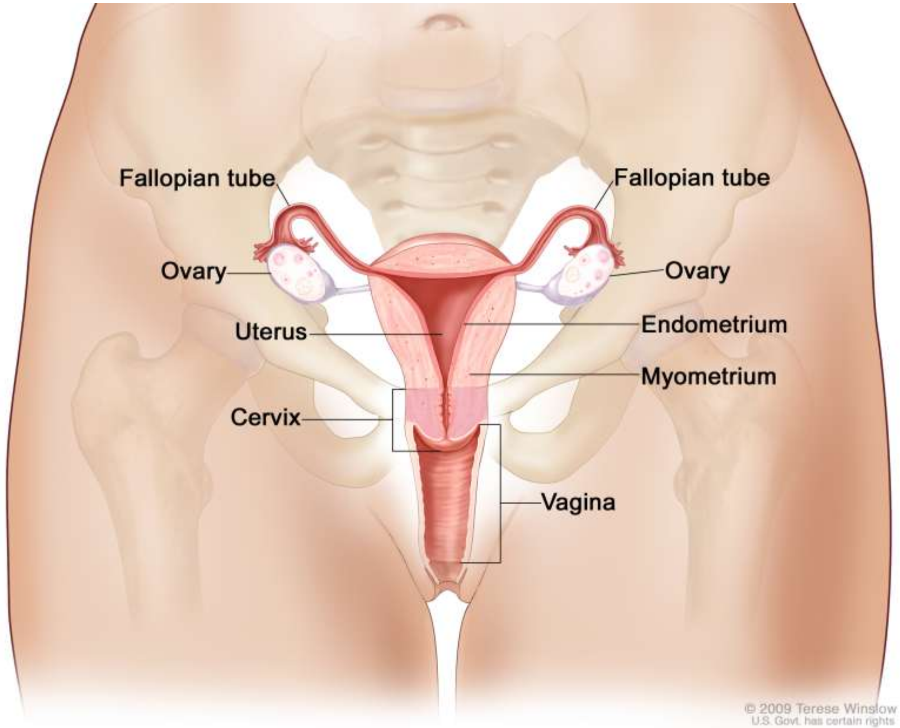


Figure: https://www.ncbi.nlm.nih.gov/books/NBK65829/

- If pregnancy results, the muscular wall of the uterus, called the **myometrium**, will contract during labor

- The outer end of the uterus is nearly closed off by the **cervix**, a ring of connective tissue that encircles a tiny opening

  – cervix holds the developing fetus in the uterus & then expands during labor, permitting passage

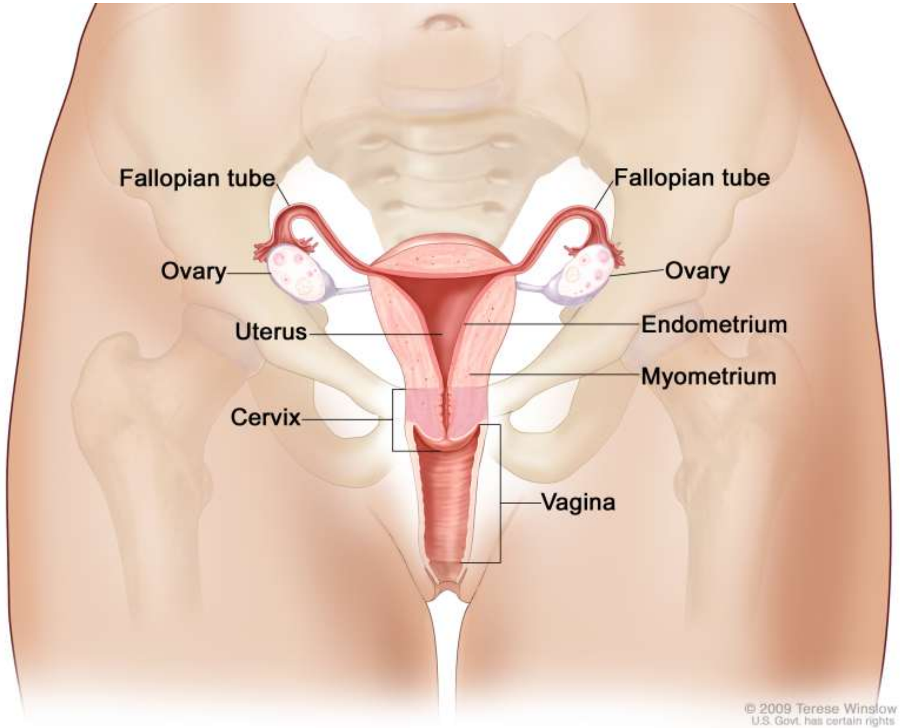
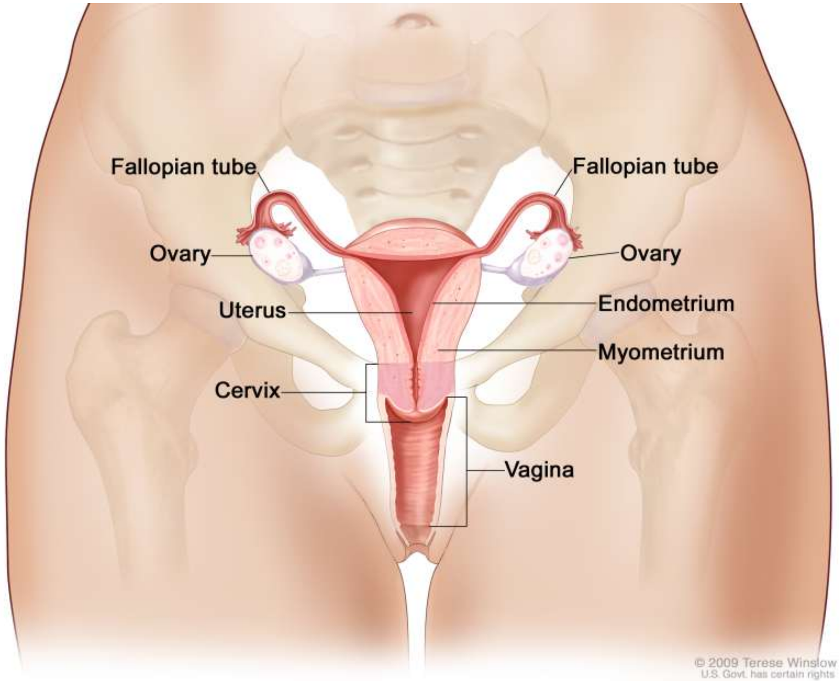


Figure: https://www.ncbi.nlm.nih.gov/books/NBK65829/

- If the egg is not fertilized, it is shed from the body, along with the endometrium, during **menstruation**
    - Small contractions of the myometrium help expel the egg & unneeded endometrial lining out through the cervix & vagina (can cause some menstrual cramping)
- Menstruation is caused by hormone level cycles



Female hormone cycles

- Female hormone levels change, following ~28 day cycle
  - Developing follicles in the ovaries secrete estrogen
    - Stimulates endometrium to become thicker & grow a large network of blood vessels & tissue

      (days 0 – 14)

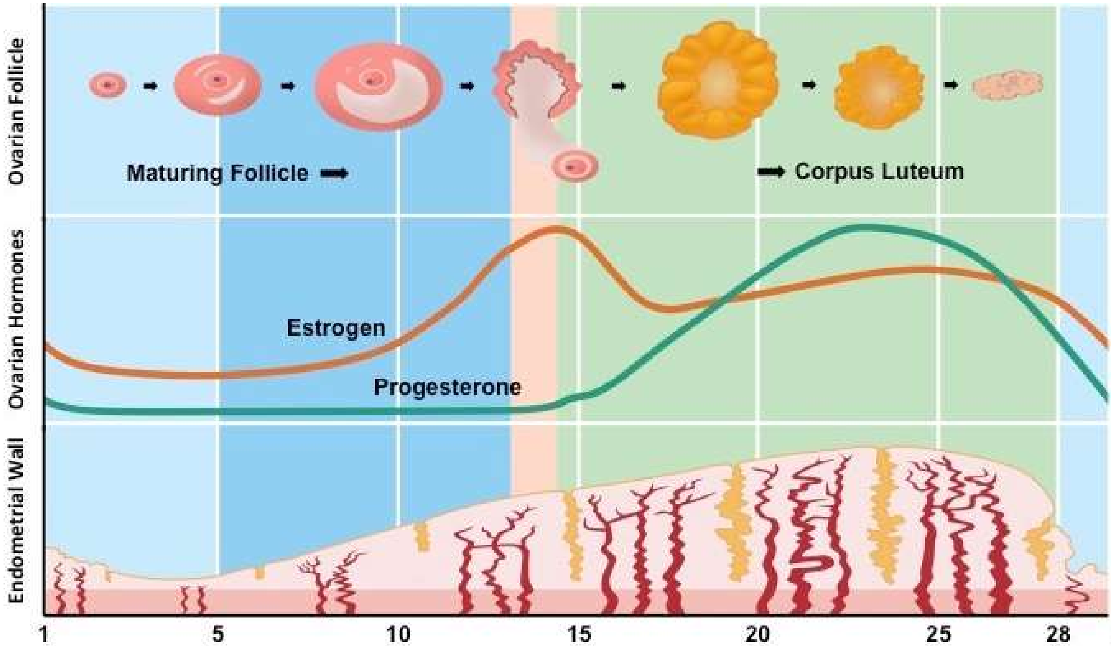


Figure: https://ib.bioninja.com.au/standard-level/topic-6-human-physiology/66-hormones-homeostasis-and/menstrual-cycle.html

- After ovulation, the corpus luteum develops & releases estrogen & progesterone – both continue to simulate the endometrium (days 14+)
  – This way, if an egg is fertilized it encounters a rich environment for growth (thick endometrium)

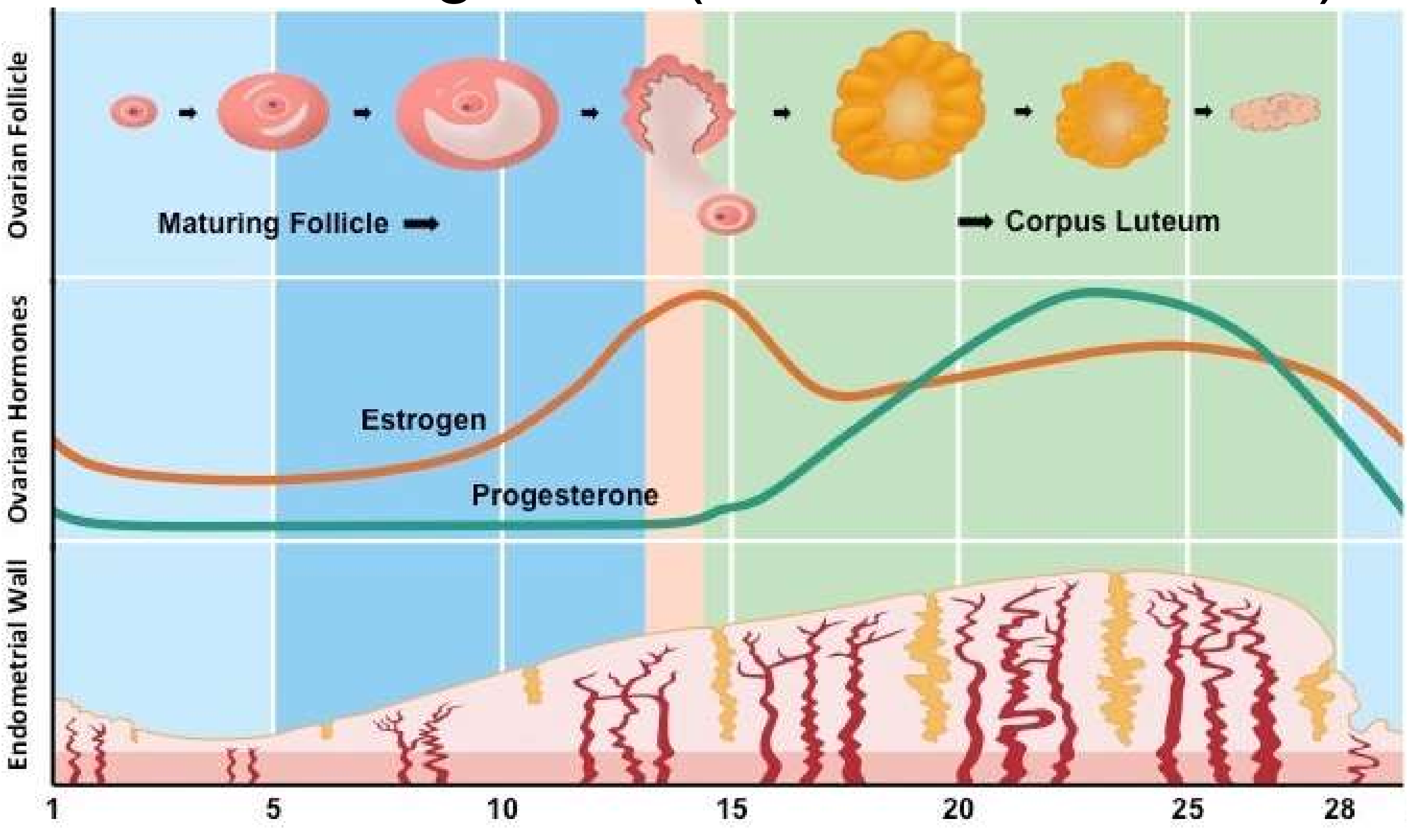
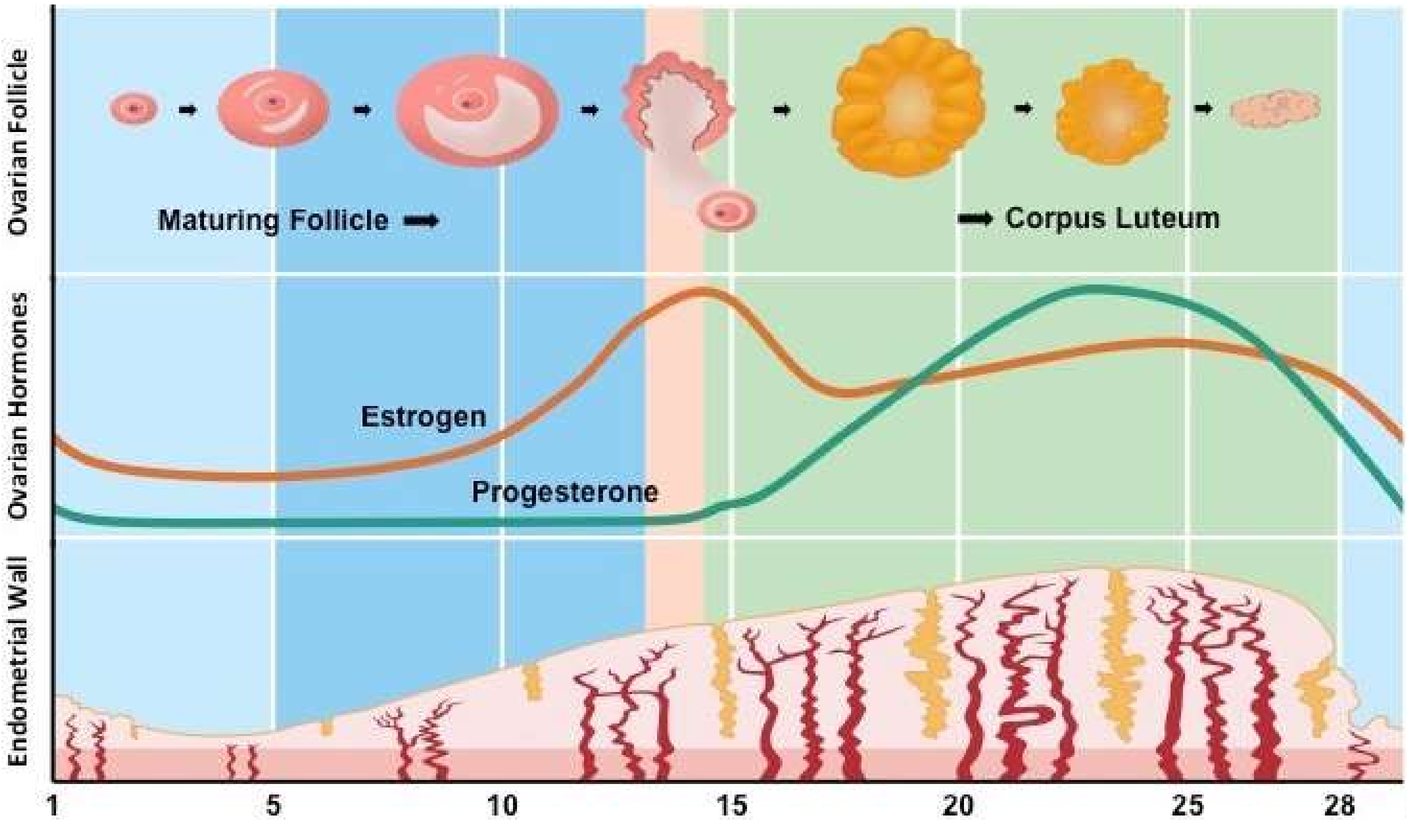


Figure: https://ib.bioninja.com.au/standard-level/topic-6-human-physiology/66-hormones-homeostasis-and/menstrual-cycle.html

- <u>If pregnancy does not occur</u>, the corpus luteum disintegrates, so estrogen & progesterone levels fall
- Drop in hormones causes the thickened endometrium to disintegrate = menstruation (day 28 to about day 5 of a new cycle)



- <u>If pregnancy does occur</u>, the corpus luteum does NOT degrade – keeps releasing estrogen & progesterone
  - Also starts releasing a 3rd hormone "CG," which circulates around body & is also excreted in urine
    - Urine strip pregnancy tests work by detecting CG (pregnant) or no CG (not pregnant)
  - During pregnancy, the endometrium develops into the placenta

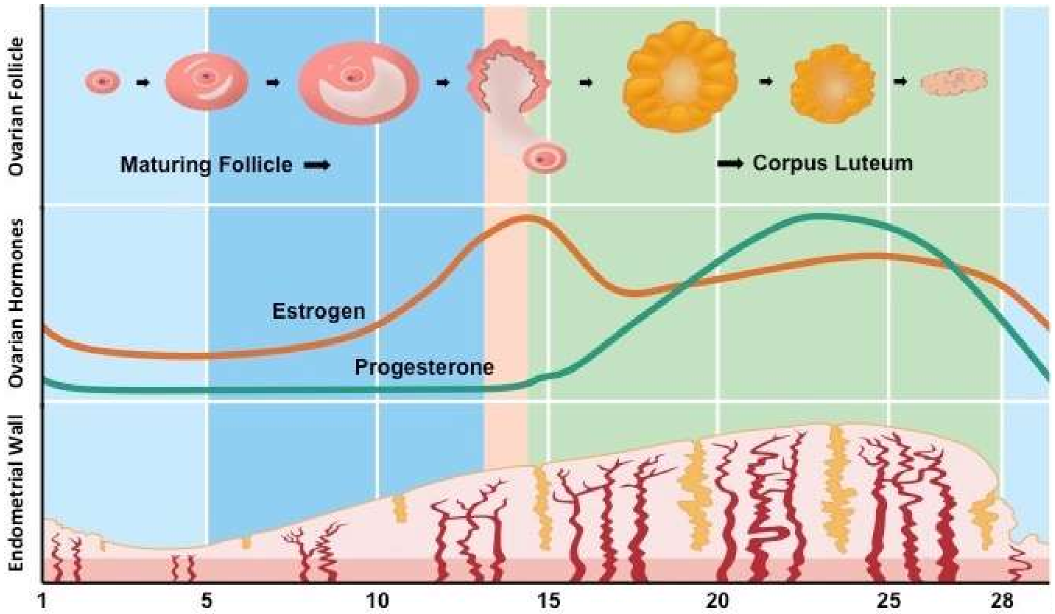


Figure: https://ib.bioninja.com.au/standard-level/topic-6-human-physiology/66-hormones-homeostasis-and/menstrual-cycle.html

Male reproductive anatomy

- <u>The male reproductive system</u>
  - Not fully contained in the abdominal cavity
  - Consists of structures that produce sperm & testosterone, ducts that conduct sperm through body, & several glands that secrete substances to nourish & support sperm

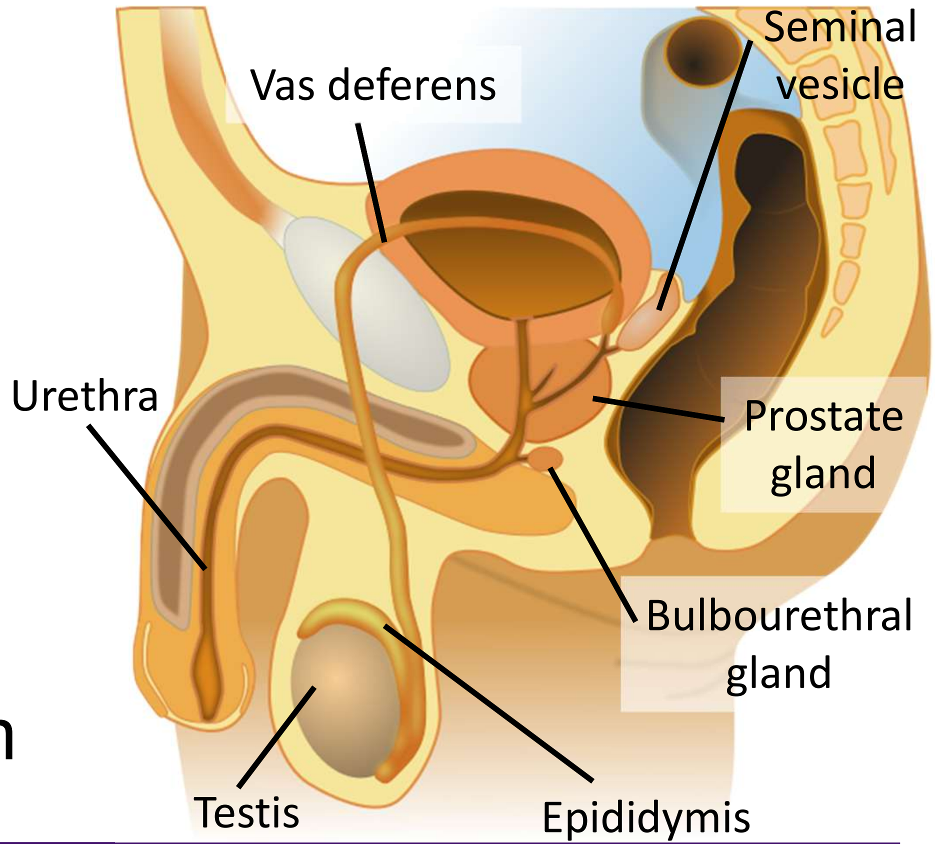


Figure: https://commons.wikimedia.org/wiki/File:Male_anatomy_blank.svg

Human male reproductive anatomy

Structure	Function in Reproduction
Testes	produce sperm & testosterone
Epididymis	store & conduct sperm
Vas deferens	conduct sperm from testes to urethra
Urethra	conducts semen (& urine) out of the body through penis
Penis	deposits sperm in female reproductive tract
Glands: Seminal vesicles Prostate Bulbourethral	secret fluid in semen to nourish & support sperm

- <u>Sperm production in the testes</u>
  - **Seminiferous tubules**
  - **Testes** are located in the scrotum: a pouch that hangs outside the main body cavity

- Keeps the testes a few degrees cooler than the body, depending on whether standing (cooler) or sitting (warmer) & type of clothing

- Cooler temps promote sperm development
  (about 100 million per day!)

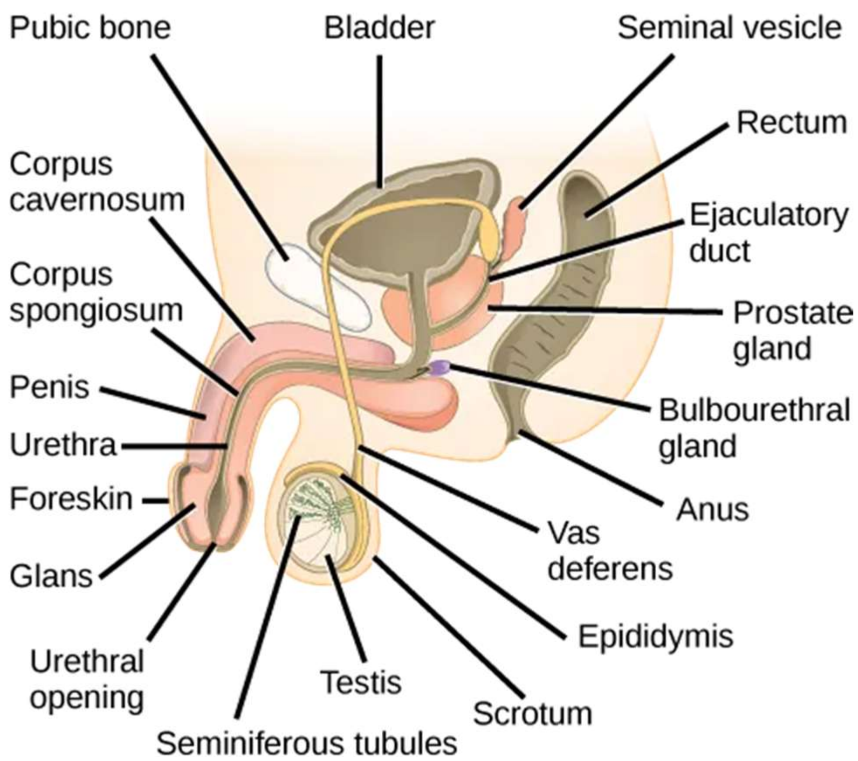
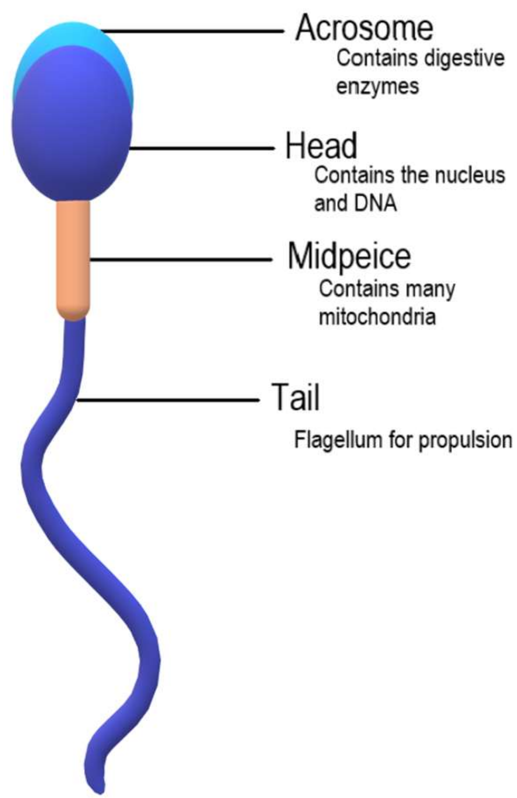


Figure: https://commons.wikimedia.org/wiki/File:Male_anatomy_blank.svg

- <u>Sperm: a very unique animal cell</u>
  - Missing most cytoplasm – the haploid nucleus nearly fills the sperm's head
  - Atop the nucleus lies a specialized structure called an **acrosome**:
    - contains enzymes to dissolve protective layers around the egg, enabling the sperm to enter & fertilize it



Males with low sperm count have trouble fertilizing eggs because they don't have enough sperm (& accompanying acrosomes) to break through the egg's barrier

Figure: https://commons.wikimedia.org/wiki/File:Sperm_Anatomy.png

- Sperm: a very unique animal cell
  - Behind the head is the midpiece, which is packed with mitochondria
    - Mitochondria provide the energy needed to move the tail (long flagellum)
  - Whiplike movements of the tail propel the sperm, allowing them to move
  - Relatively constant testosterone ensures constant new sperm growth throughout the lifespan

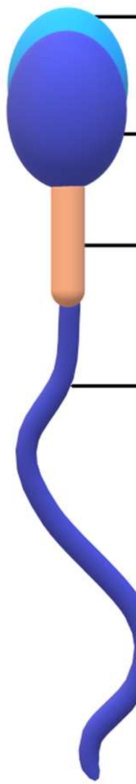


Figure: https://commons.wikimedia.org/wiki/File:Sperm_Anatomy.png

- <u>Sperm conduction & storage:</u>
  - Testes → epididymis → vas deferens → urethra
  - **Epididymis** stores sperm once they're made
  - The **vas deferens** conducts sperm to the **urethra**, which runs through the penis
  - **Urethra** conducts urine out of the body during urination & sperm out of the body during ejaculation

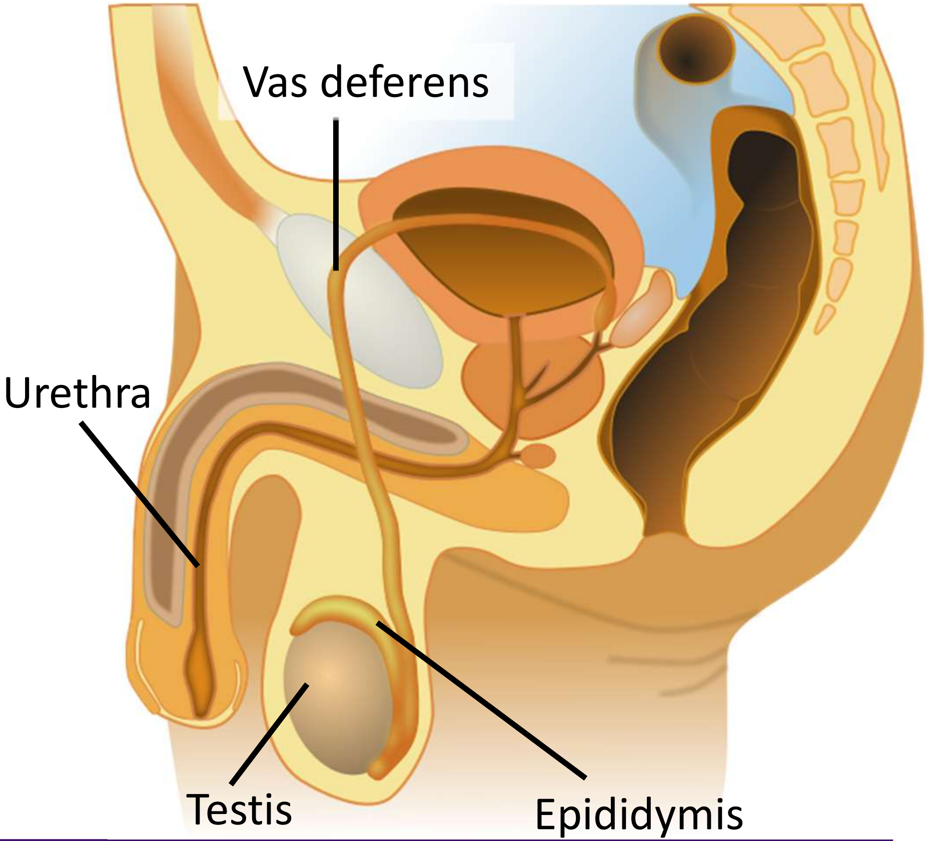
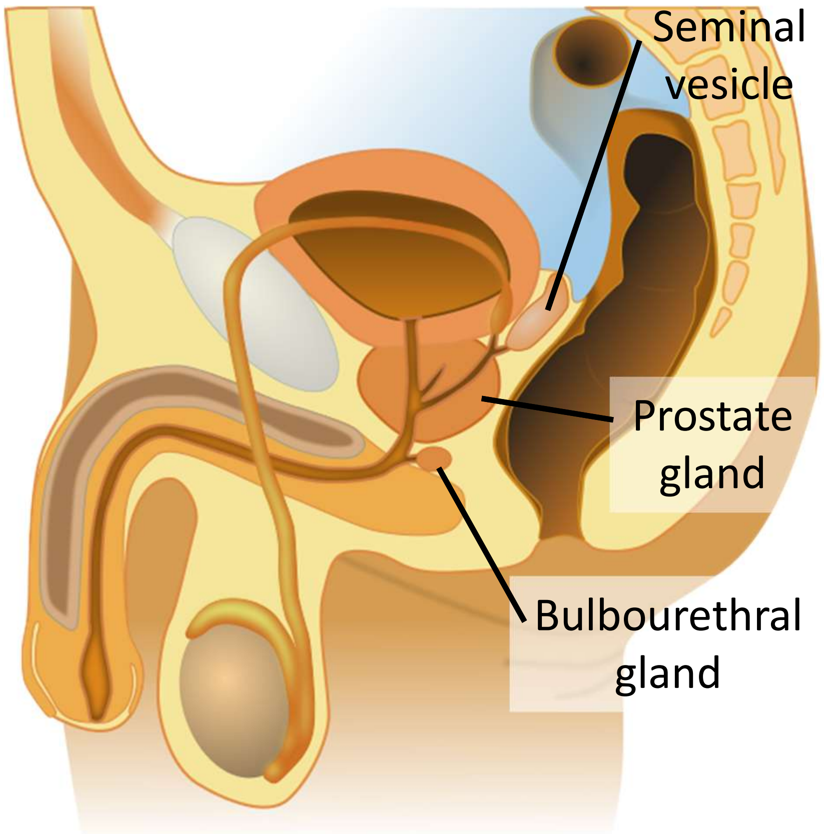


Figure: https://commons.wikimedia.org/wiki/File:Male_anatomy_blank.svg

- <u>Semen = fluid ejaculated from penis</u>
  - Only ~5% sperm, mixed with secretions from 3 different glands as it leaves the body
  - 1. The **seminal vesicles** provide fluid high in sugar, supplying mitochondria in sperm with energy, & provide a basic pH to counter the acidic vagina
    - Produce ~70% semen



- <u>Semen = fluid ejaculated from penis</u>
  - 2. The **prostate gland** provides fluid with a basic pH, & more importantly a consistency allowing sperm to move more freely
  - 3. The **bulbourethral glands** provide fluid to neutralize traces of acidic urine that may be left in the urethra

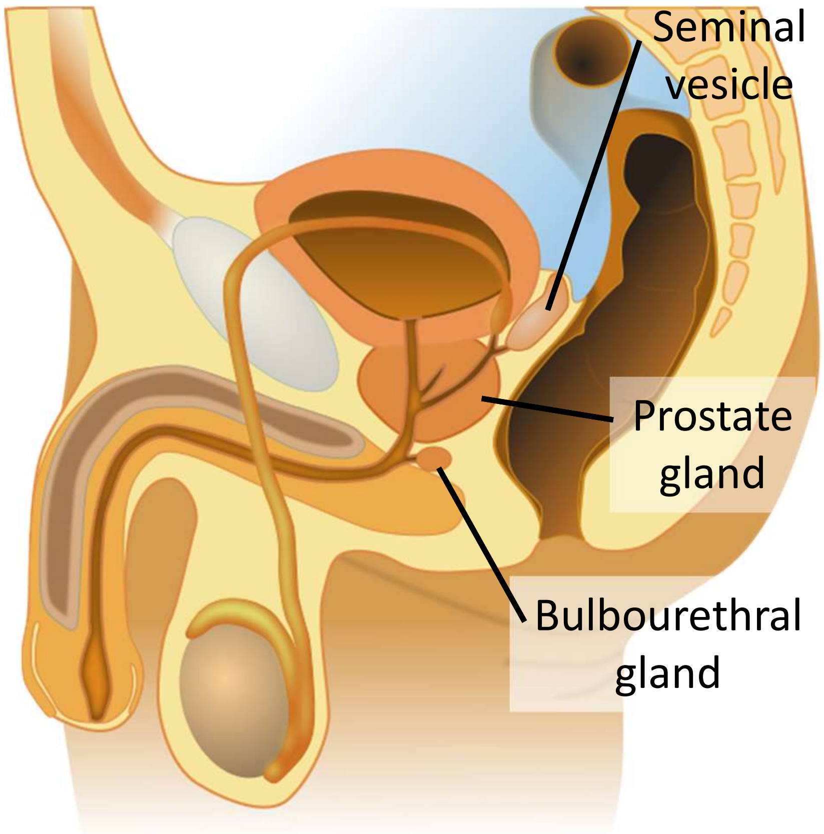
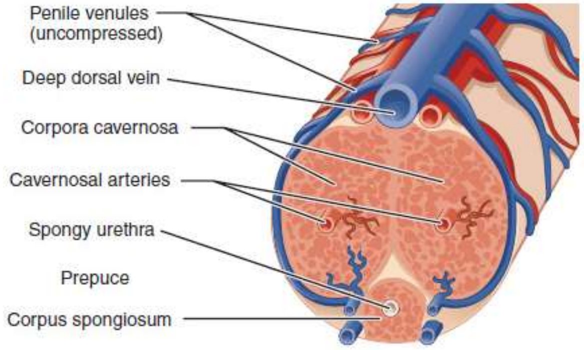


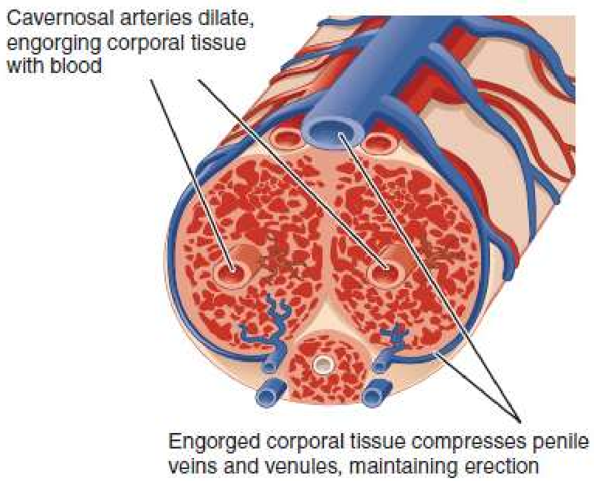
Figure: https://commons.wikimedia.org/wiki/File:Male_anatomy_blank.svg

- <u>Male role in reproductive act: erection & ejaculation</u>
  - Increased blood flow to the penis causes an **erection**, allowing for vaginal penetration
  - Increased stimulation causes **ejaculation**: muscles in the penis contract, forcing semen out through the urethra
    - Small muscle contractions before ejaculation often cause the release of some semen

Relaxed/flaccid



Erect



Increased blood through arteries (more blood from heart, engorgement of penis) causes veins to pinch (less blood leaving back to the heart) and the urethra to pinch (hard to urinate with erection)

Figure: https://commons.wikimedia.org/wiki/File:Erection_dilation.jpg

- Ejaculation can release up to hundreds of millions of sperm at once, depending on the length of time since the last ejaculation

- Once in the female reproductive tract, sperm can live for 2-5 days

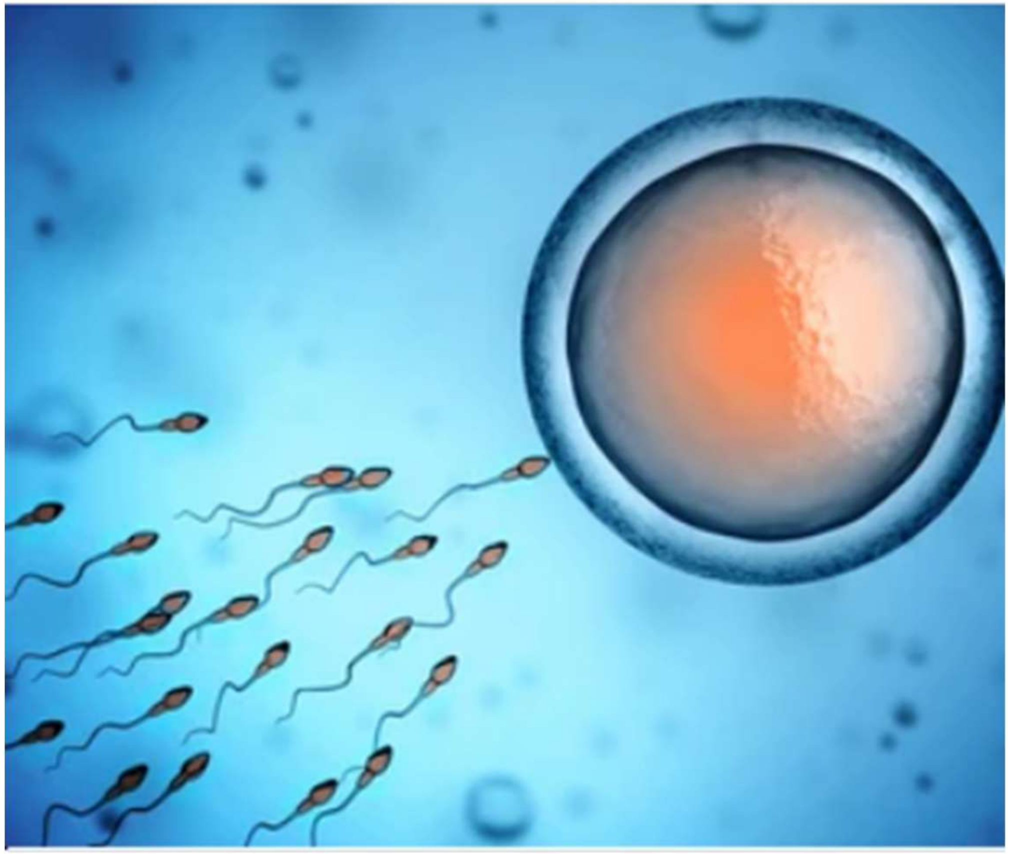
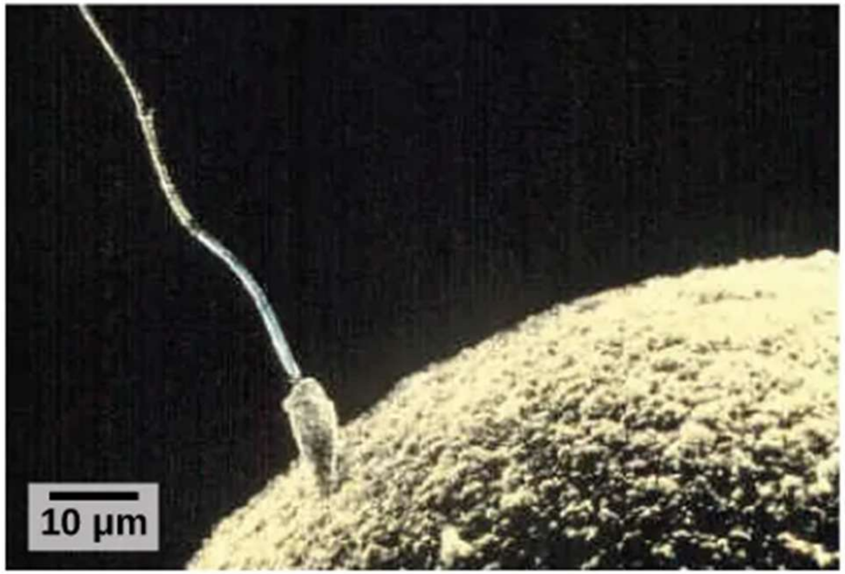
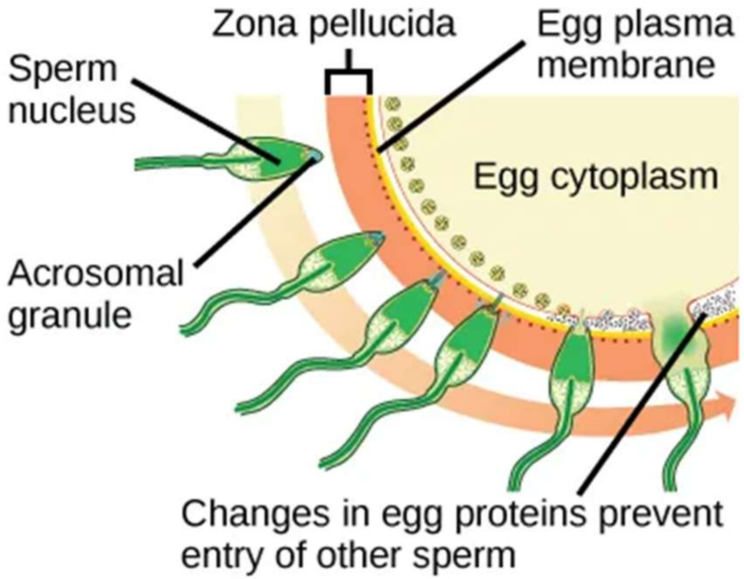


Figure: https://www.shutterstock.com/image-illustration/sperm-egg-cell-128623430



(a)



(b)

(a) Fertilization is the process in which sperm and egg fuse to form a zygote.

(b) Acrosomal reactions help the sperm degrade the glycoprotein matrix protecting the egg and allow the sperm to transfer its nucleus.

Figure: https://www.shutterstock.com/image-illustration/sperm-egg-cell-128623430

Preventing pregnancy

- If sperm meet an egg, pregnancy can occur
  - Pregnancy requires 2 things: fertilization & implantation

- Several effective techniques have been developed for **contraception** – the prevention of pregnancy
  - More permanent forms include vasectomies & tubal ligations, which disrupt the pathways in which sperm or eggs must travel
  - More temporary forms include hormone treatment, RISUG/VasalGel, & barrier methods
  - Less reliable forms include spermicide, the rhythm method, & withdrawal

- **Vasectomy**: an operation in which the vas deferens leading from each testis can be severed & tied, clamped, or sealed
  - Sperm are still produced but cannot leave the epididymis, so they are removed as waste
  - Ejaculation occurs normally with sperm-less semen
  - Reversing a vasectomy: in some cases the vas deferens can be reconnected, but it's difficult - this should only be considered as a permanent procedure

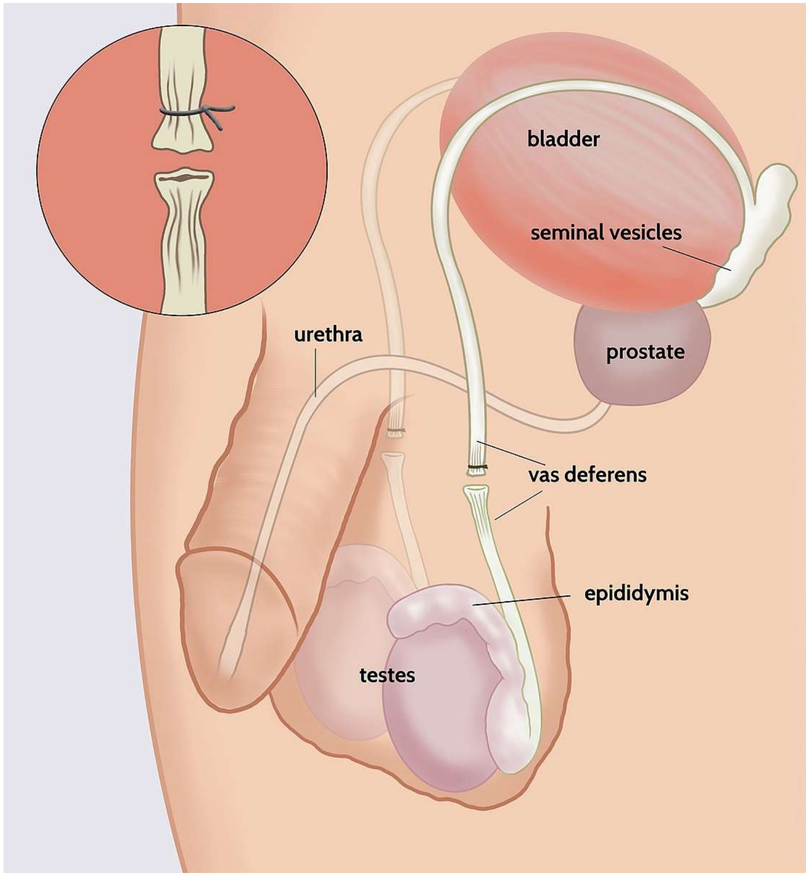


Figure: https://commons.wikimedia.org/wiki/File:Open_Vasectomy_.jpeg

– **Tubal ligation**: an operation in which the uterine tubes can be clamped or cut

- Ovulation still occurs & hormone levels do not change, but sperm cannot travel to the egg, nor can the egg reach the uterus

- Reversing tubal ligation: in some cases the uterine tubes can be reconnected, but it's difficult – should only be considered as a permanent procedure

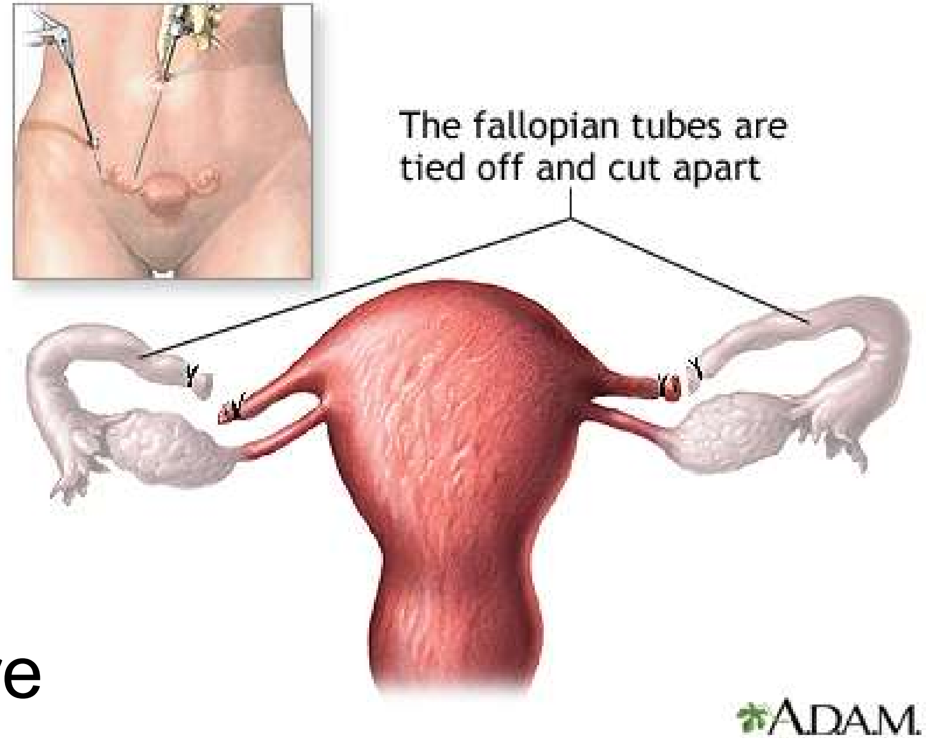


Figure: https://medlineplus.gov/ency/presentations/100044_2.htm

- Temporary birth control methods fall into 2 categories:
  - 1. Preventing ovulation
    - **Hormone treatments** such as the pill, patch, ring, or injection
      - Include just enough estrogen & progesterone to prevent ovulation
  - 2. Preventing sperm & egg from meeting
    - Barrier methods such as **condoms**, diaphragms, & sponges
    - An **intrauterine device** (IUD) is inserted into the cervix & damages sperm as they pass

- **Hormonal contraceptives:**
  - e.g. birth control pill, patch, insert (e.g. NuvaRing), or injection containing estrogen & progesterone hormones
    - Altered hormone levels in body prevent ovulation
  - When used correctly, prevent pregnancy over 99% of the time
  - Bonus: use of hormonal contraceptives reduces risk of cancer, anemia, endometriosis, acne, PMS symptoms, & ovarian cysts

Figure: https://www.factsaboutfertility.org/hormonal-contraception-linked-to-depression-informed-consent/

- **Condoms**
  – Condoms for male or female genitalia can be used to prevent egg & sperm from meeting

  – When used correctly, prevent pregnancy over 85% of the time

  – Bonus: offers some protection against STIs too

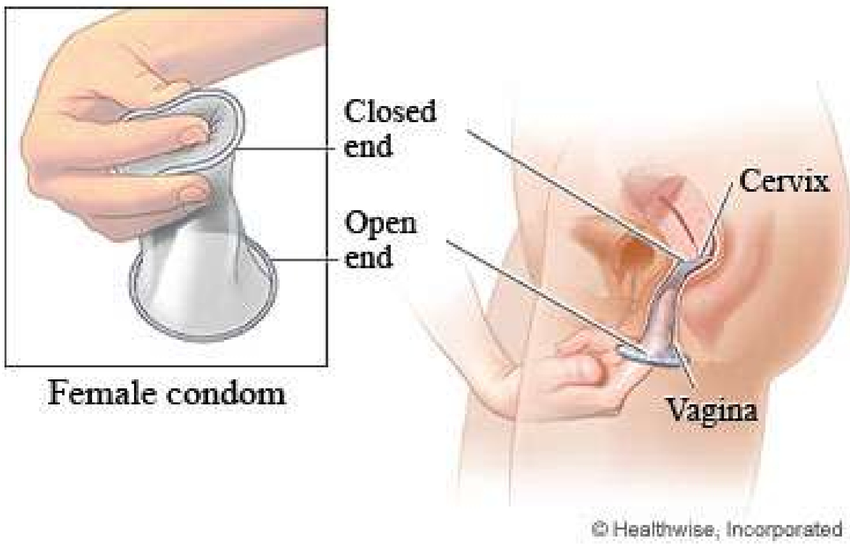


Figure: https://myhealth.alberta.ca/health/AfterCareInformation/pages/conditions.aspx?hwid=ug6452&

Male condoms that are too small may break, & those that are too big can slip off the penis. This increases the chance of pregnancy & STIs. Poor-fitting condoms can also feel uncomfortable during sex. Best fit is based on penis girth.

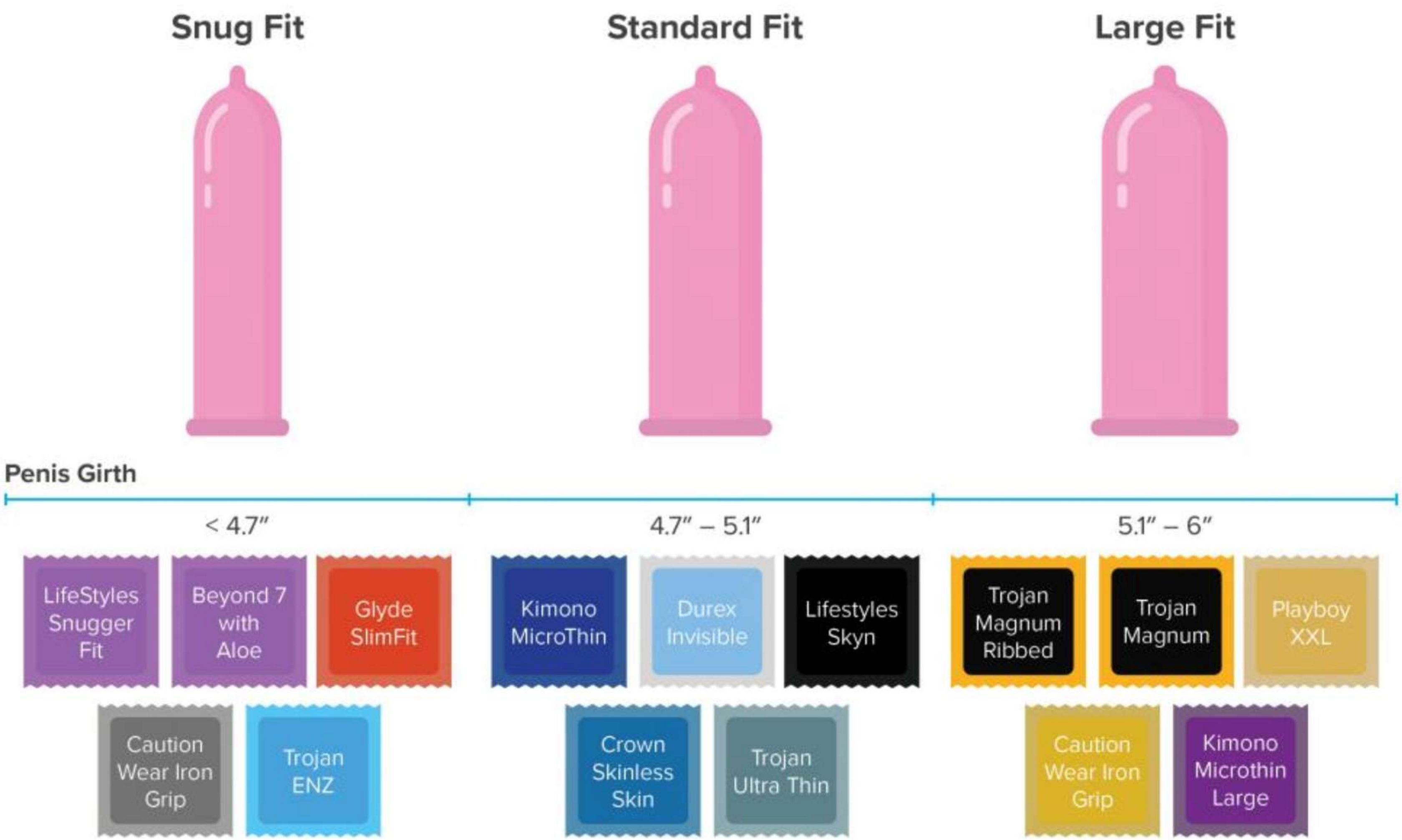
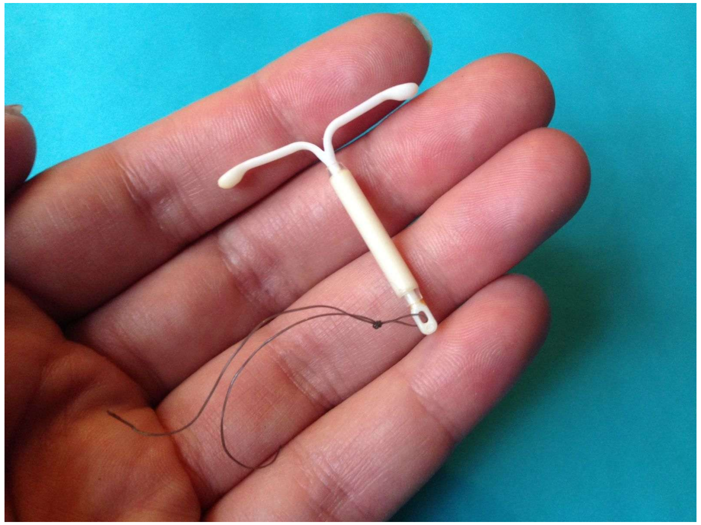


Figure: https://www.medicalnewstoday.com/articles/323707.php

- **Intrauterine devices (IUDs)**
  - Placed inside the uterus
  - Impair sperm motility & act as a spermicide
  - When used correctly, prevent pregnancy over 99% of the time
  - Bonus: one implant can last over 10 years



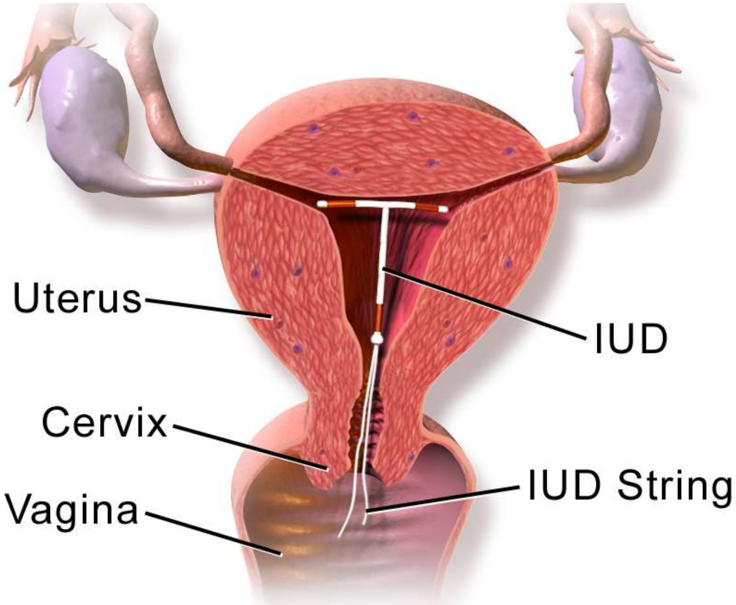
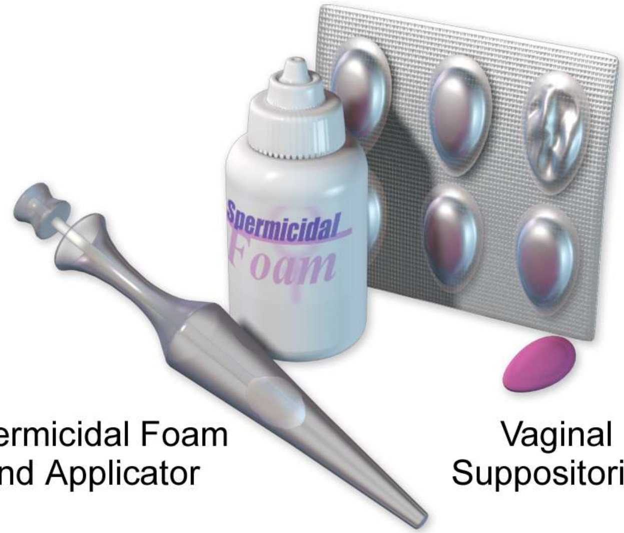


Figure: https://commons.wikimedia.org/wiki/File:Mirena_IUD_with_hand.jpg
Figure: https://commons.wikimedia.org/wiki/File:Blausen_0585_IUD.png

- Less reliable techniques
  - **Spermicide** comes as a gel, foam, or even film
    - Kills sperm on contact
    - Inserted into vagina



    - When used correctly, prevents pregnancy only ~70% of the time
    - Can be effective when paired with a condom if it is broken & a few sperm enter the vagina
    - Drawback: spermicide causes tissue irritation & can increase risk of HIV & other STIs

Figure: https://commons.wikimedia.org/wiki/File:Spermicidal_Foam_%26_Suppositories.png

- Less reliable techniques
  - **Withdrawal**: “coitus interruptus” or “pull-out method”
    - Removing the penis from the vagina just before ejaculation
    - In a “perfect-use” scenario, prevents pregnancy pretty well (~94%)
      - Actually prevents pregnancy only ~70% of the time because of pre-ejaculate (semen exit before ejaculation) and “user error”

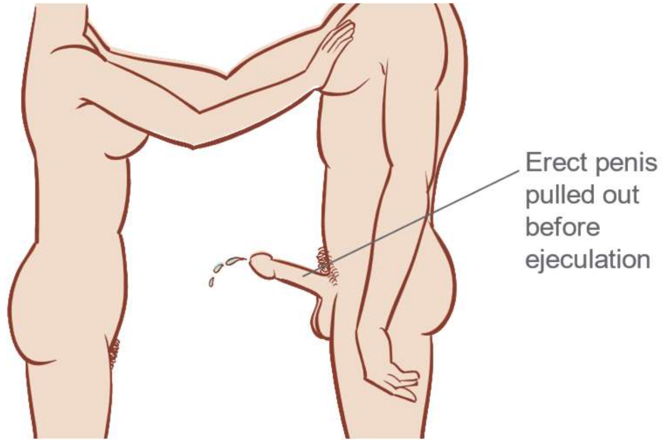


Figure: https://www.fpv.org.au/for-you/contraception/withdrawal-method-pulling-out

- Less reliable techniques
  - The **rhythm method** involves abstaining from intercourse during ovulation (most fertile time of menstrual cycle)
    - Tends to have a high failure rate because cycle varies, making ovulation difficult to predict
    - Sperm can survive for several days in the female reproductive tract, so intercourse must be avoided for several days before ovulation
      - Better for planning to get pregnant rather than preventing it

- Highly unreliable techniques:
  - **Douching:** attempting to wash sperm out of the vagina before they have entered the uterus
    - Only reduces chances of pregnancy by a maximum of 30% (70% chance of pregnancy)
      - Sperm are fast & usually reach uterus by the time douching can begin
      - The pressure of a douche can actually propel sperm into the uterus
    - Increases risk of problems during pregnancy
    - Unhealthy for vaginal environment to douche at any time

- **Emergency contraception**: pretty effective
  - The "morning after" pill contains high doses of the same hormones as in birth control pills
    - Acts primarily by preventing ovulation, but may also prevent implantation
  - Usually effective if taken within 72 hours, but ideally should be taken within 24 hours of intercourse
  - 75-90% effective depending on how long after intercourse it is administered

Figure: https://www.flickr.com/photos/jeepersmedia/16124887045

Newest forms of birth control focus on males

- Oral contraceptives are currently being researched
  - Goal is to disrupt sperm production or to alter sperm so that they can't fertilize an egg
- **VasalGel** & **intra vas devices** (IVDs) are in human testing phases
  - VasalGel: injection of a gel into the vas deferens
    - Gel hardens to the sides, allowing sperm to pass but destroying them as they do
  - IVD: blocks vas deferens so that sperm cannot pass
  - Both = safe, non-toxic, completely reversible, & can last over 10 years

Infections transmitted through sexual interaction

- Some microbes move from host to host during sexual encounters, causing **sexually transmitted infections (STIs)** – *sometimes known as STDs*
  - Can also be passed via shared needles, childbirth, or breastfeeding
- Most contraception methods do NOT protect against STIs
  - Condoms limit physical contact of the genitals, *reducing* STI exposure (but not preventing fully)
  - Other ways to *reduce* spread = vaccinations (for some STIs) & reducing # of sexual partners

- STIs can be caused by <u>bacteria</u>, <u>viruses</u>, or <u>protists</u> (protozoa) that infect the reproductive tract or organs

Common STDs

STD	Organism Type					Transmission	
	Bacterial	Viral	Protozoa	Can be cured	Cannot be cured, but can be managed	Fluids	Skin to Skin
Chlamydia	X			X		X	
Gonorrhea	X			X		X	
Syphilis	X			X			X
Trichomoniasis			X	X		X	
HSV (herpes simplex virus)		X			X		X
HPV (Human Papilloma Virus)		X			X		X

Figure: http://actforyouth.net/sexual_health/behaviors/std.cfm

- **Bacterial STIS**
  - Common symptoms = painful urination & discharge, lesions, fever, pregnancy complications, etc.
    - Can = infertility in females if untreated for long
  - Treated with antibiotics (curable)
    - However, some antibiotic-resistant strains have been discovered
  - Many people show no symptoms
    - e.g. Chlamydia often referred to as the “Silent Epidemic” because some males & ~70% of females show no symptoms at all

- **Viral STIS**
  - There are many different viruses, including HPV, HIV/AIDS, & genital herpes – no cures for viruses

- **HIV/AIDS**: "Human Immunodeficiency Virus / Acquired Immunodeficiency Syndrome"
  - Causes immune deficiency & is eventually fatal
    - Asymptomatic for months & even many years
      - Many don't know they're infected
  - No cure but some drug combinations can slow the progression and can keep the disease under control for many years, especially if caught early
    - AIDS "functionally" cured?

- **Genital herpes:**
  - Common symptoms = painful, burning, or itching blisters on the genitals
    - Outbreaks can recur throughout lifetime, though often occur less frequently & are less severe over many years
  - Pretty common: ~16% of American adults
    - Though many will never show symptoms
  - No cure, but antiviral drugs can help reduce the frequency & severity of blister outbreaks

Figure: https://en.wikipedia.org/wiki/File:SOA-Herpes-genitalis-female.jpg

Figure: https://en.wikipedia.org/wiki/File:SOA-Herpes-genitalis-male.jpg

- **Human papillomavirus (HPV):** many types/strains
  - Some can cause cervical & other forms of cancer
    - A vaccine against the cancer-causing strains is available for everyone but doesn't cure existing infections
  - Some can cause genital or anal warts
  - Most types never cause any symptoms
    - ~80% of sexually active individuals are infected with at least 1 strain at some point in their life

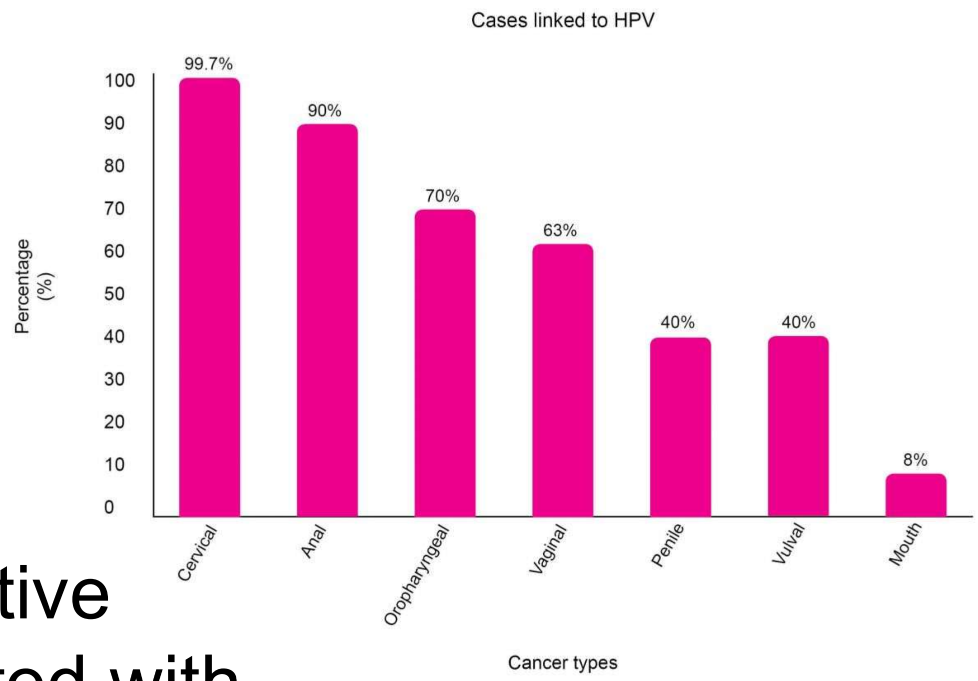


Figure: https://www.jostrust.org.uk/information/hpv/hpv-other-cancers